



5 The continuous random variable X has cumulative distribution function F given by

$$F(x) = \begin{cases} 0 & x < 2, \\ \frac{(x-2)^2}{12} & 2 \leq x < 4, \\ 1 - \frac{(8-x)^2}{24} & 4 \leq x \leq 8, \\ 1 & x > 8. \end{cases}$$

(a) Sketch the graph of the probability density function of X . [3]

(b) Find $E(X)$. [3]

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